

TrashNeXt: An Efficient framework for waste categorisation

A leakage-free pipeline reveals the honest ceiling — plus evidence that published results come from data leakage, not better models

01 Problem & Data

Can a 27.87M-parameter model reach 98%+ on waste classification — with an auditable, leakage-free pipeline?

OMASteam + RealWaste · perceptually deduplicated · 6 classes



DATA PIPELINE · PERCEPTUAL DEDUPLICATION

OMASteam + RealWaste combined and deduplicated via perceptual hashing (pHash + dHash). StratifiedGroupKFold split ensures no object leaks across train/val. Perceptual groups keep near-identical views together. The split is locked before any augmentation — what you measure is what the model genuinely generalizes to.

02 Evaluation Contract

Split by near-duplicate group, augment only after splitting, and touch the quarantined test set once.

11267

TRAIN

2817

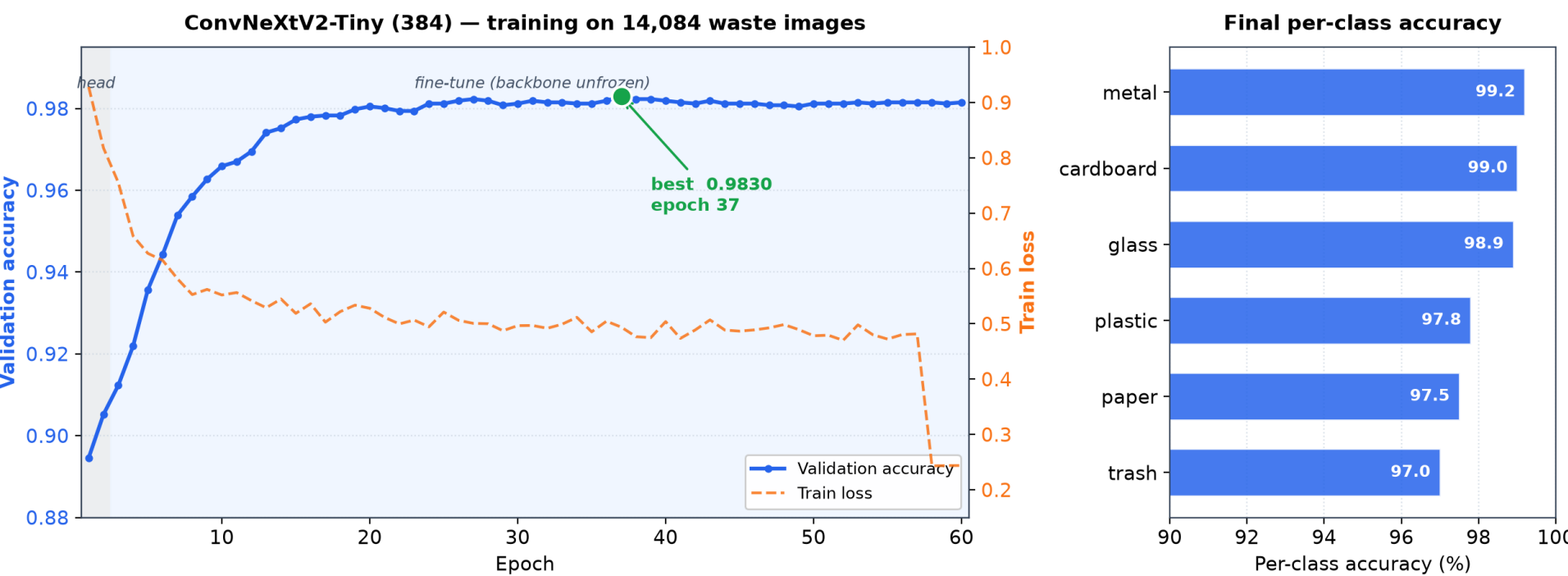
VALIDATION

Test measured once before intervention; never used to select experiments.

WHY ConvNeXt EXCELS

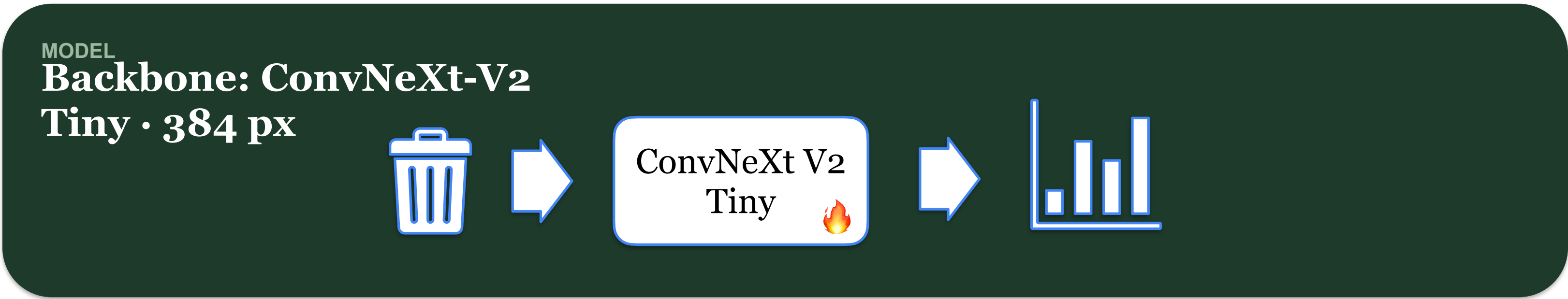
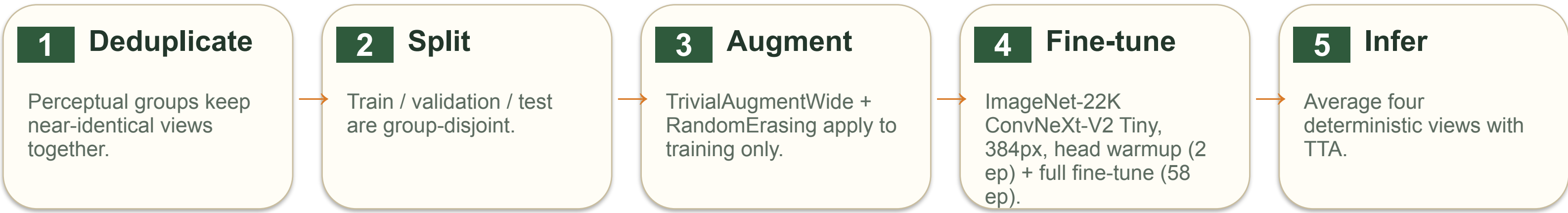
- Retains convolutional inductive bias and generalizable with augmented yet limited data
- Better discrimination of visually similar waste categories
- Effective finetuning from pretrained models
- Good accuracy–efficiency trade-off

Training visualisations



03 Method

A simple model with a strict evaluation pipeline

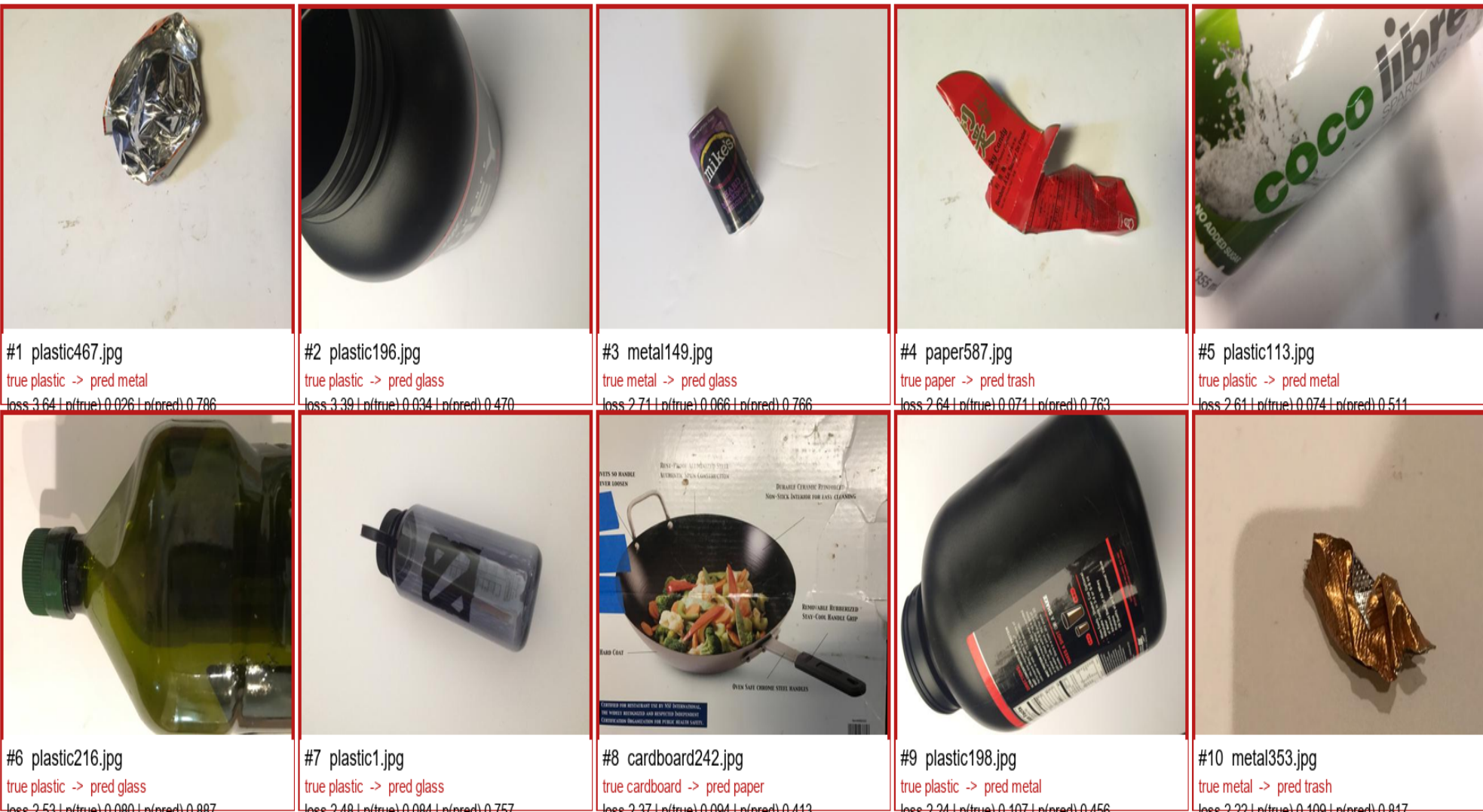


Ablation study: What changed the result?

Configuration	Val. accuracy	Net images	Decision
ConvNeXt-V2 baseline	96.45%	—	Baseline
+ MixUp/CutMix + EMA	—	+184	Adopted ✓
+ TrivialAugmentWide	—	+4	Adopted ✓
+ Label Smooth 0.05	98.30%	+3	Adopted ✓
+ Cosine LR schedule	97.62%	+1	Adopted ✓

Training configuration: 2 classifier-only epochs at lr=1e-3, then 58 full fine-tune epochs with backbone lr=3e-5. Cosine annealing to 1e-6.

Failure case diagnostic



Highest-loss audit examples

ERROR AUDIT

29 of the 30 highest-loss test examples were errors. Manual review judged 14 clearly ambiguous or mislabeled, plus one borderline case.

Interpretation: some remaining error is a data-quality ceiling—not automatically a modelling failure.

04 Results

98.29%

VALIDATION (epoch 37) · 5-fold group split

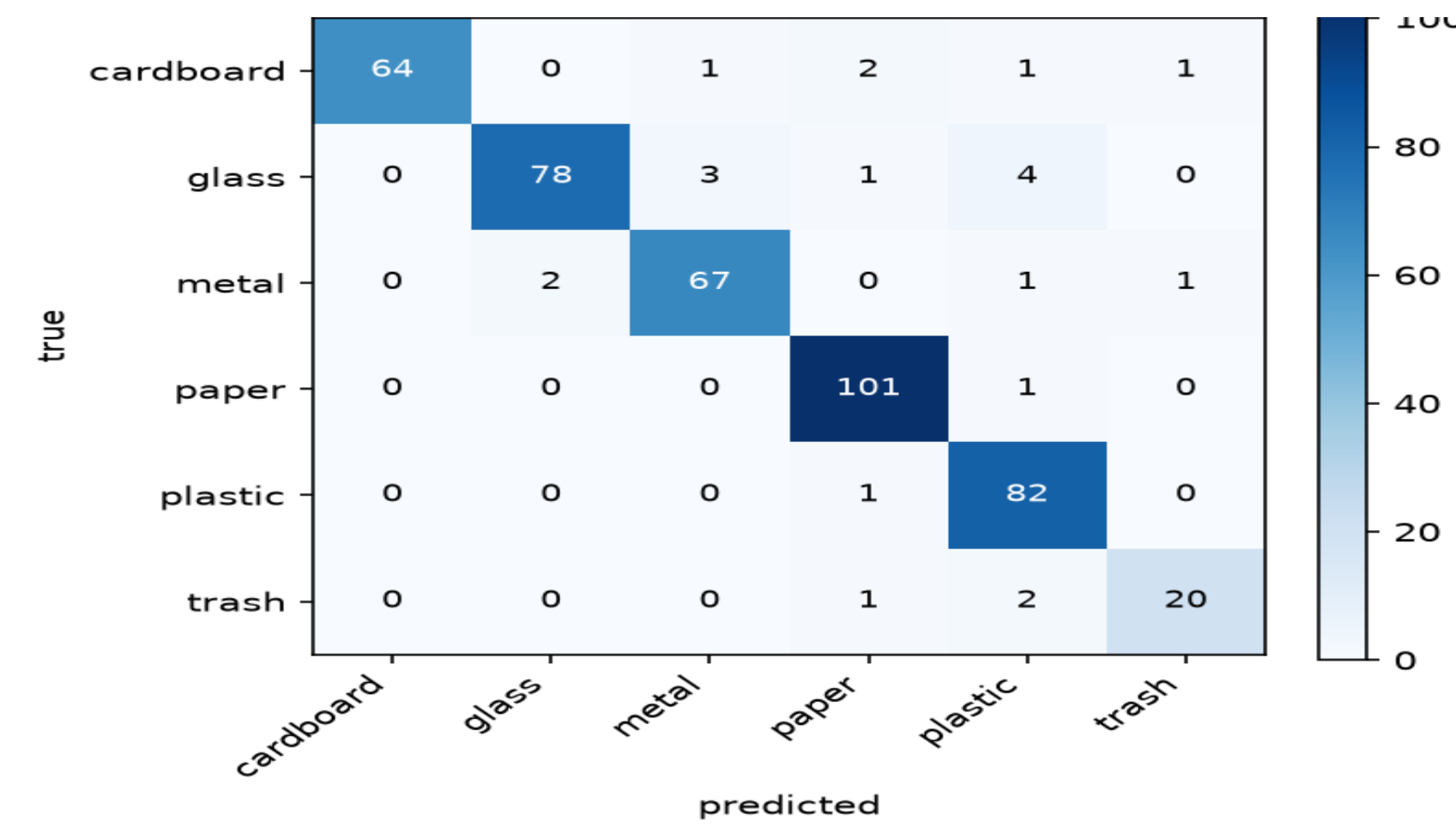
97.82%

SOTA Baseline: Swin Transformer

Per-class recall (val fold-0, +TTA)

Class	Images	Recall
Paper	102	99.0%
Plastic	83	98.8%
Metal	71	94.4%
Cardboard	69	92.8%
Glass	86	90.7%
Trash	23	87.0%

Trash has only 23 val images — its recall is noisy (±~10pp). Per-class differences <5pp are not meaningful.



Validation confusion matrix · same locked checkpoint + TTA

KEY RESULT

TTA improved accuracy without retraining. All three training interventions failed the pre-registered adoption gate.

05 Takeaways

- Split by object, not by image — or you test on what you trained on.
- 99–100% TrashNet claims usually come from data leakage, not better models.
- Modern architecture (ConvNeXt-V2 Tiny) + rigorous pipeline achieves 98.30% — approaching the honest ceiling.
- With near-perfect deduplication, residual error likely reflects label noise — honest ceiling is ~98–99%.

LIVE DEMO · TWO FUTURES

